Supplementary information:
When high flux helps single-pixel imaging — a contrast–dead-time operating map

This supplement collects the complete tables and the one derivation that support the main text. Section, equation, and table numbers carry an "S" prefix; references of the form "main-text Section X," "Eq. (main X)" point to the primary manuscript. Every table is generated by a committed script under `code/round63/figs/` (named in each caption/comment) that reads the frozen campaign artifact and prints the tabulated numbers; no value is imputed, and empty or non-numeric cells are dropped from every statistic rather than filled.

S1 Why a full heteroscedastic Gaussian likelihood fails at high load

The reconstruction of the main text (Eq. (main 2)) uses the renewal quasi-likelihood (RQL) data term rather than a full heteroscedastic Gaussian likelihood in the integrated count. This section shows why the Gaussian model is not merely less accurate but is *non-coercive* in the high-load regime, which is the rigorous form of the brightness-inflation pathology that RQL removes by construction.

Consider modelling the single-frame count N_i of a non-paralyzable detector (dead time τ, window $T = \nu\tau$, load $\rho_i = \lambda_i \tau$) as a Gaussian with load-dependent mean and variance, $N_i \sim \mathcal{N}\big(\mu(\rho_i), \sigma^2(\rho_i)\big)$. The per-frame negative log-likelihood is

$$\ell_i(\lambda_i) \;=\; \tfrac{1}{2}\log\sigma^2(\rho_i) \;+\; \frac{\big(N_i - \mu(\rho_i)\big)^2}{2\,\sigma^2(\rho_i)} \;+\; \text{const}, \tag{S1}$$

in which the first ("log-det") term is the heteroscedastic normalization. For the active-start renewal count, each registered event costs a deterministic dead time τ followed by an exponential wait $\mathrm{Exp}(\lambda)$, so the inter-event time has mean $(1+\rho)/\lambda$ and variance $1/\lambda^2$. Renewal theory then gives, to leading order in ν,

$$\mu(\rho) \;=\; \nu\,\frac{\rho}{1+\rho}, \qquad \sigma^2(\rho) \;=\; \nu\,\frac{\rho}{(1+\rho)^3}. \tag{S2}$$

The mean is monotone in ρ, but the variance is not: differentiating the log-det term,

$$\frac{\partial}{\partial\rho}\,\tfrac{1}{2}\log\sigma^2(\rho) \;=\; \frac{1}{2}\left(\frac{1}{\rho} - \frac{3}{1+\rho}\right) \;=\; \frac{1-2\rho}{2\,\rho(1+\rho)}. \tag{S3}$$

The sign of this "log-det incentive" *reverses at* $\rho = \frac{1}{2}$: the modelled count variance rises with load for $\rho < \frac{1}{2}$ and falls for $\rho > \frac{1}{2}$, peaking at the mid-load point where roughly half of each slot is occupied by dead time. Below $\rho = \frac{1}{2}$ the normalization penalizes an inflated fitted rate; *above* $\rho = \frac{1}{2}$ it rewards one, because a larger λ shrinks σ^2 and drives $\frac{1}{2}\log\sigma^2 \to -\infty$.

This is decisive at ceiling counts. When a frame saturates, $N_i\tau \geq T$ (equivalently $N_i \geq \nu$), the count can no longer discriminate among large rates, and the plug-in variance $\sigma^2(\rho) \to 0$ as $\rho \to \infty$ while the squared-residual term stays bounded. Objective (S1) is then *unbounded below* in λ_i: the Gaussian estimator can lower its loss without limit by inflating the brightness of any saturated measurement, so the maximum-likelihood/MAP problem is non-coercive and its minimizer is ill-posed. The exhibited heteroscedastic-Gaussian pseudo-likelihood is noncoercive along the displayed high-load direction, and the tested TV settings did not remove the resulting failure. Sufficiently coercive penalties can bound such a direction, so no universal claim is made over all convex regularizers.

RQL removes the pathology at its source. Its data term (main-text Eq. (main 2)), $\frac{1}{M}\sum_i \big[(T - N_i\tau)\lambda_i(x) - N_i \log\lambda_i(x)\big]$, retains only the renewal *mean* relationship—its single-measurement stationary point $\hat{\lambda} = N/(T -$

$N\tau$) is exactly the classical dead-time rate correction—and carries no $\frac{1}{2}\log\sigma^2$ term. It is convex and coercive in λ (and, through the affine map, in $x \geq 0$); the only genuine identifiability boundary that survives is the exact-count ceiling $N\tau \geq T$, at which the objective is correctly reported as unbounded and the ceiling fraction is recorded per cell rather than silently fitted. The convex surrogate thus keeps the correct mean response while discarding the variance-collapse term that makes the full Gaussian likelihood fail deep in the dead-time regime; that the surrogate costs essentially nothing in reconstruction quality is verified against the exact likelihood in Section S4.

S2 Full NATURAL-30 cohort

Table S1 lists the complete per-image acquisition-speed endpoint for the preregistered natural-image cohort (secondary, no gate; the same dense-pattern protocol as Study 1), from which the main text reports only the cohort summary and one image pair. The cohort spans two synthetic stress scenes: `low_contrast` is the only strongly negative rung ($\Delta Q = -1.78$ dB) and `fine_lines` sits essentially at zero ($\Delta Q = +0.07$ dB), bracketing the smooth-scene low-benefit regime.

Table S1: Full NATURAL-30 cohort (secondary, no gate): per-image acquisition-speed endpoint on the RQL arm, reproduced verbatim from the frozen analyzer output. SRU = SAFE_RANGE_UNINFORMATIVE ($S_{\text{gate}} = 1$ by the frozen censoring taxonomy). $T_{\text{safe}}, T_{\text{fast}}$ are Q_{90} crossing times (s); R_j the safe-arm quality span; ΔQ_{budget} the fixed-dwell radiometric-PSNR gain at $\nu = 2000$. The two synthetic stress scenes bracket the cohort: `low_contrast` ($\Delta Q = -1.78$ dB, the only strongly negative rung) and `fine_lines` ($\Delta Q = +0.07$ dB). Cohort median $S_{\text{gate}} = 5.70$; median $\Delta Q = +1.44$ dB, positive in 29/30 scenes; 12/30 scenes safe-range-uninformative.

image	family	status	S_{gate}	T_{safe}	T_{fast}	R_j	Q_{90}	ΔQ	seeds
fine_lines	fine_lines	SRU	1.000	–	–	0.09	–	0.07	5
gray_staircase	gray_staircase	NORMAL	4.939	0.313	0.06338	2.73	12.16	3.11	5
low_contrast	low_contrast	SRU	1.000	–	–	0.00	–	-1.78	5
sparse_dots	sparse_dots	NORMAL	7.645	0.3155	0.04128	6.16	26.14	3.00	5
stl_00	stl_00	NORMAL	5.991	0.37	0.06177	0.77	13.22	1.25	5
stl_01	stl_01	SRU	1.000	–	–	0.10	–	0.29	5
stl_02	stl_02	SRU	1.000	–	–	0.02	–	0.86	5
stl_03	stl_03	NORMAL	5.711	0.3447	0.06036	1.33	16.23	1.25	5
stl_04	stl_04	SRU	1.000	–	–	0.30	–	1.42	5
stl_05	stl_05	NORMAL	7.209	0.3621	0.05023	0.57	14.77	1.25	5
stl_06	stl_06	NORMAL	6.987	0.3568	0.05107	0.93	13.23	1.51	5
stl_07	stl_07	NORMAL	6.761	0.344	0.05088	1.91	13.71	2.57	5
stl_08	stl_08	NORMAL	10.916	0.2978	0.02728	1.59	12.80	2.46	5
stl_09	stl_09	NORMAL	6.289	0.3677	0.05846	0.94	15.77	2.30	5
stl_10	stl_10	SRU	1.000	–	–	0.00	–	0.37	5
stl_11	stl_11	SRU	1.000	–	–	0.41	–	1.14	5
stl_12	stl_12	SRU	1.000	–	–	0.43	–	1.03	5
stl_13	stl_13	SRU	1.000	–	–	0.28	–	1.35	5
stl_14	stl_14	NORMAL	6.617	0.3131	0.04732	0.94	14.49	1.62	5
stl_15	stl_15	NORMAL	7.804	0.3301	0.0423	0.80	11.89	2.26	5
stl_16	stl_16	NORMAL	6.980	0.3547	0.05081	1.07	13.80	1.46	5
stl_17	stl_17	NORMAL	8.375	0.3617	0.04319	0.53	13.00	1.58	5
stl_18	stl_18	NORMAL	5.295	0.2284	0.04313	1.02	14.47	1.85	5
stl_19	stl_19	NORMAL	6.568	0.3557	0.05416	1.73	12.18	2.33	5
stl_20	stl_20	NORMAL	5.692	0.3342	0.05871	1.15	14.67	1.28	5
stl_21	stl_21	SRU	1.000	–	–	0.34	–	0.77	5
stl_22	stl_22	SRU	1.000	–	–	0.36	–	1.63	5
stl_23	stl_23	NORMAL	7.803	0.3186	0.04083	1.03	12.52	1.74	5
text	text	NORMAL	7.834	0.3393	0.04331	0.77	13.77	2.02	5
usaf_bars	usaf_bars	SRU	1.000	–	–	0.38	–	0.95	5

S3 Complete detector-mismatch tables

The main text reports six anchor perturbations at the preregistered operating point; the anchor subset is reproduced in Table S2 and the complete variant sweep in Table S3. The full table adds the severity sweeps (dead-time error to ±20%, dark rate 5–50%, afterpulsing to 10%) and the start-mode, guard-time and dead-time-jitter axes.

Table S2: Detector-mismatch anchor subset at the preregistered anchor ($M/n = 0.5$, $\nu = 500$, 64^2), paired against the matched no-mismatch baseline. Entries are median paired radiometric-PSNR shifts (dB); the RQL flux bias is the variant's own median radiometric flux deviation at the fast point. PRECORRECT dPSNR is reported at the fast point only: at the safe point PRECORRECT is a degenerate floored reconstruction (flux_dev $= -1.0$ for all rows, identical to its own baseline), so its safe dPSNR is trivially ~ 0 and uninformative. The paralyzable-data / non-paralyzable-model anchor is not instantiated in the pilot and is reported MISSING rather than imputed.

variant	RQL safe (dB)	RQL fast (dB)	RQL flux bias (fast)	PRECORRECT fast (dB)
tau_err +10%	-0.00	-0.10	0.06	-0.50
tau_err -10%	0.00	-0.12	-0.06	0.45
dark 10% (known)	-0.05	0.16	0.00	-0.04
dark 10% (unknown)	-0.34	-0.30	0.10	-0.79
afterpulsing 2%	-0.00	-0.00	0.01	-0.12
afterpulsing 5%	-0.01	-0.09	0.03	-0.23
paralyzable-data / non-paralyzable-model	MISSING	MISSING	MISSING	MISSING

Three honesty notes are carried explicitly: (i) at the safe point PRECORRECT is a degenerate floored reconstruction (flux_dev $= -1.0$, identical to its own baseline), so its safe dPSNR is uninformative and is omitted from the anchor columns; (ii) the dark50%-known PRECORRECT safe row is an *un-flooring anomaly*—a known 50% dark rate lifts PRECORRECT off its floor (flux_dev $\to 0$) and yields a -2.60 dB paired shift, the single large PRECORRECT swing in the table; and (iii) the paralyzable-data / non-paralyzable-model anchor is not instantiated in the pilot and is reported MISSING rather than imputed.

Table S3: Complete detector-mismatch variant table: every (variant, arm, $\bar{\rho}$) group at the $M/n = 0.5$, $\nu = 500$ anchor, including the severity sweeps (dead-time error to $\pm 20\%$, dark rate 5–50%, afterpulsing to 10%) and the start-mode, guard-time and dead-time-jitter axes. n = rows in group; n_{pair} = per-(image,seed) pairs used for the paired dPSNR; Δflux is the difference of median radiometric flux deviations (variant − baseline). MISSING marks $\bar{\rho} = 0.3$ interaction cells with no matched baseline. The dark50%-known PRECORRECT safe row is the un-flooring anomaly: the normally floored PRECORRECT safe reconstruction (flux_dev $= -1.0$) un-floors to flux_dev ≈ 0 under a known 50% dark rate, giving a -2.60 dB paired shift. All values verbatim from the frozen artifact; non-numeric PSNR dropped from medians, never imputed.

variant	arm	$\bar{\rho}$	med. PSNR	med. flux	n	dPSNR	method	n_{pair}	Δflux
baseline	RQL	0.05	14.53	-0.00	36	0.00	baseline	36	0.00
baseline	RQL	0.6	14.95	0.00	36	0.00	baseline	36	0.00
baseline	RQL	1	14.70	0.00	36	0.00	baseline	36	0.00
baseline	PRECORRECT	0.05	6.36	-1.00	36	0.00	baseline	36	0.00
baseline	PRECORRECT	0.6	4.17	0.00	36	0.00	baseline	36	0.00
baseline	PRECORRECT	1	4.33	0.00	36	0.00	baseline	36	0.00
baseline	POISSON-LIN	0.05	14.47	-0.05	36	0.00	baseline	36	0.00
baseline	SAT-POISSON	0.05	14.52	-0.00	36	0.00	baseline	36	0.00
baseline	POISSON-LIN	0.6	11.95	-0.37	36	0.00	baseline	36	0.00
baseline	SAT-POISSON	0.6	14.82	0.00	36	0.00	baseline	36	0.00
baseline	POISSON-LIN	1	10.82	-0.50	36	0.00	baseline	36	0.00
baseline	SAT-POISSON	1	14.89	0.00	36	0.00	baseline	36	0.00
tau_err=+0.1	RQL	0.05	14.52	0.00	36	-0.00	paired	36	0.01
tau_err=+0.1	RQL	0.3	14.59	0.03	36	MISSING	no-baseline	0	MISSING
tau_err=+0.1	RQL	0.6	14.79	0.06	36	-0.10	paired	36	0.06
tau_err=+0.1	RQL	1	14.46	0.11	36	-0.25	paired	36	0.11
tau_err=+0.1	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
tau_err=+0.1	PRECORRECT	0.3	3.74	0.03	36	MISSING	no-baseline	0	MISSING
tau_err=+0.1	PRECORRECT	0.6	3.68	0.06	36	-0.50	paired	36	0.06
tau_err=+0.1	PRECORRECT	1	3.41	0.11	36	-0.87	paired	36	0.11
tau_err=+0.2	RQL	0.05	14.53	0.01	36	0.00	paired	36	0.01
tau_err=+0.2	RQL	0.6	14.30	0.14	36	-0.46	paired	36	0.14
tau_err=+0.2	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00

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variant	arm	$\bar{\rho}$	med. PSNR	med. flux	n	dPSNR	method	n_{pair}	Δflux
tau_err=+0.2	PRECORRECT	0.6	3.12	0.14	36	-1.06	paired	36	0.14
tau_err=-0.1	RQL	0.05	14.52	-0.01	36	0.00	paired	36	-0.00
tau_err=-0.1	RQL	0.3	14.59	-0.03	36	MISSING	no-baseline	0	MISSING
tau_err=-0.1	RQL	0.6	14.84	-0.06	36	-0.12	paired	36	-0.06
tau_err=-0.1	RQL	1	14.61	-0.09	36	-0.27	paired	36	-0.09
tau_err=-0.1	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
tau_err=-0.1	PRECORRECT	0.3	4.23	-0.03	36	MISSING	no-baseline	0	MISSING
tau_err=-0.1	PRECORRECT	0.6	4.61	-0.06	36	0.45	paired	36	-0.06
tau_err=-0.1	PRECORRECT	1	5.11	-0.09	36	0.73	paired	36	-0.09
tau_err=-0.2	RQL	0.05	14.52	-0.01	36	-0.00	paired	36	-0.01
tau_err=-0.2	RQL	0.6	14.53	-0.11	36	-0.34	paired	36	-0.11
tau_err=-0.2	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
tau_err=-0.2	PRECORRECT	0.6	5.00	-0.11	36	0.85	paired	36	-0.11
dark10%_known	RQL	0.05	14.34	-0.00	36	-0.05	paired	36	-0.00
dark10%_known	RQL	0.6	14.87	0.00	36	0.16	paired	36	-0.00
dark10%_known	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
dark10%_known	PRECORRECT	0.6	4.18	0.00	36	-0.04	paired	36	-0.00
dark10%_unknown	RQL	0.05	14.17	0.10	36	-0.34	paired	36	0.10
dark10%_unknown	RQL	0.6	14.47	0.10	36	-0.30	paired	36	0.10
dark10%_unknown	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
dark10%_unknown	PRECORRECT	0.6	3.40	0.10	36	-0.79	paired	36	0.10
dark25%_known	RQL	0.05	14.41	-0.00	36	0.04	paired	36	-0.00
dark25%_known	RQL	0.6	14.98	0.00	36	0.02	paired	36	0.00
dark25%_known	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
dark25%_known	PRECORRECT	0.6	4.06	0.00	36	-0.11	paired	36	0.00
dark5%_known	RQL	0.05	14.41	-0.00	36	0.07	paired	36	-0.00
dark5%_known	RQL	0.6	15.06	0.00	36	0.08	paired	36	-0.00
dark5%_known	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
dark5%_known	PRECORRECT	0.6	4.24	0.00	36	0.05	paired	36	-0.00
dark50%_known	RQL	0.05	14.40	-0.00	36	-0.03	paired	36	0.00
dark50%_known	RQL	0.6	14.54	0.00	36	-0.24	paired	36	0.00
dark50%_known	PRECORRECT	0.05	3.68	-0.00	36	-2.60	paired	36	1.00
dark50%_known	PRECORRECT	0.6	3.86	0.00	36	-0.26	paired	36	0.00
p_ap=0.01	RQL	0.05	14.45	0.01	36	0.06	paired	36	0.01
p_ap=0.01	RQL	0.6	15.33	0.01	36	0.15	paired	36	0.01
p_ap=0.01	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
p_ap=0.01	PRECORRECT	0.6	4.17	0.01	36	-0.04	paired	36	0.01
p_ap=0.02	RQL	0.05	14.47	0.02	36	-0.00	paired	36	0.02
p_ap=0.02	RQL	0.3	14.71	0.02	36	MISSING	no-baseline	0	MISSING
p_ap=0.02	RQL	0.6	14.92	0.01	36	-0.00	paired	36	0.01
p_ap=0.02	RQL	1	14.93	0.01	36	0.09	paired	36	0.01
p_ap=0.02	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
p_ap=0.02	PRECORRECT	0.3	3.89	0.01	36	MISSING	no-baseline	0	MISSING
p_ap=0.02	PRECORRECT	0.6	4.04	0.01	36	-0.12	paired	36	0.01
p_ap=0.02	PRECORRECT	1	4.32	0.01	36	-0.09	paired	36	0.01
p_ap=0.05	RQL	0.05	14.40	0.05	36	-0.01	paired	36	0.05
p_ap=0.05	RQL	0.6	14.80	0.03	36	-0.09	paired	36	0.03
p_ap=0.05	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
p_ap=0.05	PRECORRECT	0.6	3.98	0.03	36	-0.23	paired	36	0.03
p_ap=0.1	RQL	0.05	14.37	0.09	36	-0.19	paired	36	0.09
p_ap=0.1	RQL	0.3	14.45	0.08	36	MISSING	no-baseline	0	MISSING
p_ap=0.1	RQL	0.6	14.78	0.06	36	-0.16	paired	36	0.06
p_ap=0.1	RQL	1	15.05	0.05	36	0.03	paired	36	0.05
p_ap=0.1	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
p_ap=0.1	PRECORRECT	0.3	3.51	0.08	36	MISSING	no-baseline	0	MISSING
p_ap=0.1	PRECORRECT	0.6	3.84	0.06	36	-0.45	paired	36	0.06
p_ap=0.1	PRECORRECT	1	4.02	0.05	36	-0.41	paired	36	0.05
start=delayed	RQL	0.05	14.51	-0.00	36	-0.00	paired	36	-0.00
start=delayed	RQL	0.6	14.96	-0.00	36	-0.00	paired	36	-0.00
start=delayed	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
start=delayed	PRECORRECT	0.6	4.18	-0.00	36	0.03	paired	36	-0.00
guard=2.5e-07	RQL	0.05	14.53	-0.00	36	0.00	paired	36	0.00
guard=2.5e-07	RQL	0.6	14.95	0.00	36	0.00	paired	36	0.00
guard=2.5e-07	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
guard=2.5e-07	PRECORRECT	0.6	4.17	0.00	36	0.00	paired	36	0.00
guard=5e-08	RQL	0.05	14.53	-0.00	36	0.00	paired	36	0.00
guard=5e-08	RQL	0.6	14.95	0.00	36	0.00	paired	36	0.00

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variant	arm	$\bar{\rho}$	med. PSNR	med. flux	n	dPSNR	method	n_{pair}	Δflux
guard=5e-08	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
guard=5e-08	PRECORRECT	0.6	4.17	0.00	36	0.00	paired	36	0.00
jitter_cv=0.05	RQL	0.05	14.35	0.00	36	0.06	paired	36	0.00
jitter_cv=0.05	RQL	0.6	14.81	0.00	36	0.01	paired	36	0.00
jitter_cv=0.05	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
jitter_cv=0.05	PRECORRECT	0.6	4.31	0.00	36	0.01	paired	36	0.00
jitter_cv=0.1	RQL	0.05	14.34	0.00	36	0.07	paired	36	0.00
jitter_cv=0.1	RQL	0.6	14.89	0.00	36	-0.02	paired	36	0.00
jitter_cv=0.1	PRECORRECT	0.05	6.36	-1.00	36	0.00	paired	36	0.00
jitter_cv=0.1	PRECORRECT	0.6	4.33	0.00	36	0.01	paired	36	0.00

S4 Complete exact-likelihood-versus-RQL comparison

Because RQL is a convex surrogate for the exact renewal likelihood, the main text verifies at 8^2 and 16^2 that it tracks the exact likelihood under an identical TV functional and the identical frozen selection rule. Table S4 gives the complete 12×4 grid of the median exact-minus-RQL radiometric-PSNR gap and the pair-level statistics over all 864 paired reconstructions. The median gap is negligible throughout the deployment zone and grows only into the extreme-load corner; individual cells differ in both directions (up to 9.23 dB), consistent with the nonconvexity of the exact objective and with no systematic advantage to either arm.

S5 No-gate occupancy-ladder tables

Table S5 gives the per-image fixed-dwell high-flux gain on every rung of the frozen occupancy ladder (per-image seed-mean radiometric PSNR at $\bar{\rho} = 0.6$ minus $\bar{\rho} = 0.05$, both at $\nu = 2000$, under equal mean detector load and equal incident dose across occupancies). The gate-carrying $k = 16$ column is the frozen ΔQ_{budget} from the Study-2 analyzer; the $k = 512/32/1$ columns are the no-gate control and robustness rungs. The cohort medians (+0.64, +4.13, +1.16, +3.56 dB at $k = 512, 32, 16, 1$) are nonmonotone in sparsity—$k = 32$ attains the largest median gain—exposing the contrast–conditioning trade-off discussed in the main text.

S6 Dynamic-scattering detail

Table S6 details the dynamic-scattering secondary (multiplicative lognormal AR(1) factors, $\text{CV}(\alpha) = 0.20$, applied before dead time; $k = 16$ geometry, six frozen mechanism representatives), giving the mean and sample sd of the RQL radiometric PSNR at each $(\bar{\rho}, \nu)$ and the per-ν high-flux gain. The gain persists and grows with dwell (+0.66, +1.84, +2.32 dB at $\nu = 20, 200, 2000$).

S7 Ensemble and undersampling sensitivity (S2B/S2C)

Tables S7 and S8 report the supplementary geometry/ensemble sensitivity: the undersampling budget $M \in \{1024, 2048\}$ for dense Bernoulli(1/2) illumination (S2B), and the complementary Hadamard-type pairs (`hadpair`) and gamma-modulated patterns (`gam4`) at $M = 2048$ (S2C). Entries are median RQL radiometric PSNR at $\bar{\rho} \in \{0.05, 0.6, 1.0\}$ and $\nu \in \{100, 500, 2000\}$. Consistent with the multiplex-limited reading of the dense regime, the RQL quality is stable across sampling budgets and pattern ensembles and rises modestly with load at long dwell. (Provenance note: the S2C pilot table is a regenerated supplementary artifact; its fragment records its provenance separately from the frozen campaign outputs. The R19 correction route applies only to the paper-2 elapsed-time endpoint.)

S8 Descriptive measurement-audit distributions

Table S9 summarizes the descriptive measurement audit on the RQL arm of the 64^2 detail cohort: the refit-bootstrap discrepancy-ratio d_{ratio} quantiles and the leak-suspect counts by $(\bar{\rho}, \nu)$, with the overall audit-status tally in the caption. Two caveats from the main text are repeated here. The audit is descriptive and is *not* a model-adequacy test: every audit statistic is excluded from regularization selection, reconstruction, cell inclusion,

Table S4: Complete EXACT-TV versus RQL comparison on the frozen $8^2/16^2$ grid (864 paired reconstructions). Cells give the median radiometric-PSNR gap PSNR(EXACT-TV) − PSNR(RQL) in dB per (side, $\bar{\rho}, \nu$) (pair count in parentheses; three seeds × six scenes per cell). The median gap stays within 0.011 dB in magnitude throughout the deployment zone ($\bar{\rho} \leq 1$) and reaches only −0.05 dB in the extreme-load corner (16^2, $\bar{\rho} \geq 0.6$, $\nu = 2000$). Over all pairs: 141 exactly identical, median +0.00 dB, maximum |gap| = 9.23 dB, and only 9.0% of pairs exceed 0.1 dB in either direction – consistent with the nonconvexity of the exact objective and with no systematic advantage to either arm.

side	$\bar{\rho}$	ν=20	ν=100	ν=500	ν=2000
8^2	0.03	+0.000	-0.000	+0.000	+0.000
	0.10	-0.003	+0.000	+0.000	+0.000
	0.30	-0.001	-0.000	+0.000	-0.000
	0.60	+0.007	+0.000	+0.000	+0.000
	1.00	-0.005	-0.002	-0.001	-0.001
	2.00	-0.003	-0.001	-0.001	-0.001
16^2	0.03	-0.000	+0.000	+0.000	+0.000
	0.10	+0.001	+0.000	+0.000	+0.000
	0.30	-0.001	-0.000	-0.000	-0.011
	0.60	-0.001	+0.000	-0.000	-0.053
	1.00	+0.009	+0.000	-0.001	-0.054
	2.00	+0.012	-0.001	-0.011	-0.026

and all confirmatory pass/fail decisions. Its refit-bootstrap reference distribution is conditional on a smoothed plug-in scene, so the empirical tail ranks are not calibrated model-class p-values and can be anti-conservative in the upper tail when the true scene is rougher than the plug-in—the widening upper d_{ratio} quantiles at long dwell and high load reflect this plug-in conditioning, not a detected model failure.

Table S9: Descriptive measurement-audit distributions on the RQL arm of the 64^2 detail cohort (5400 audited cells; audit_status = OK for all 5400/5400). Per $(\bar{\rho}, \nu)$: the refit-bootstrap discrepancy-ratio d_{ratio} quantiles and the count of leak-suspect cells (True in 30 of 5400 cells overall). *The audit is descriptive and is not a model-adequacy test:* all audit statistics are excluded from regularization selection, reconstruction, cell inclusion, and every confirmatory pass/fail decision. The reference distribution is conditional on a smoothed plug-in scene, so its tail ranks are not calibrated model-class p-values and can be anti-conservative when the true scene is rougher than the plug-in.

$\bar{\rho}$	ν	n	$d_{5\%}$	$d_{50\%}$	$d_{95\%}$	$d_{\max}$	leak
0.05	5	120	0.873	0.988	1.125	1.195	4
0.05	10	120	0.883	0.988	1.117	1.205	2
0.05	20	120	0.895	0.993	1.109	1.185	2
0.05	50	120	0.914	0.993	1.088	1.148	2
0.05	100	120	0.925	0.998	1.089	1.119	1
0.05	200	120	0.928	1.007	1.072	1.109	0
0.05	500	120	0.922	1.010	1.084	1.127	3
0.05	1000	120	0.948	1.008	1.101	1.172	1
0.05	2000	120	0.945	1.025	1.141	1.261	0
0.30	5	120	0.929	0.994	1.065	1.111	0
0.30	10	120	0.918	1.004	1.074	1.134	3
0.30	20	120	0.921	1.000	1.070	1.119	0
0.30	50	120	0.935	1.006	1.103	1.161	1
0.30	100	120	0.942	1.021	1.091	1.113	0
0.30	200	120	0.944	1.023	1.107	1.177	0
0.30	500	120	0.967	1.045	1.156	1.253	0
0.30	1000	120	0.971	1.073	1.230	1.337	0
0.30	2000	120	0.984	1.133	1.575	1.699	0
0.60	5	120	0.914	0.996	1.045	1.090	0
0.60	10	120	0.926	0.995	1.081	1.131	1

continued on next page

$\bar{\rho}$	ν	n	$d_{5\%}$	$d_{50\%}$	$d_{95\%}$	$d_{\max}$	leak
0.60	20	120	0.929	0.999	1.080	1.121	1
0.60	50	120	0.939	1.013	1.084	1.137	0
0.60	100	120	0.948	1.020	1.073	1.142	0
0.60	200	120	0.961	1.034	1.124	1.192	0
0.60	500	120	0.955	1.077	1.286	1.335	0
0.60	1000	120	0.962	1.076	1.392	1.511	0
0.60	2000	120	1.001	1.134	1.556	1.666	0
1.00	5	120	0.913	0.974	1.048	1.088	0
1.00	10	120	0.928	1.019	1.079	1.160	3
1.00	20	120	0.926	1.010	1.095	1.146	1
1.00	50	120	0.939	1.008	1.095	1.145	0
1.00	100	120	0.953	1.029	1.141	1.176	0
1.00	200	120	0.950	1.040	1.182	1.243	1
1.00	500	120	0.962	1.078	1.296	1.367	0
1.00	1000	120	0.970	1.099	1.402	1.475	0
1.00	2000	120	1.042	1.200	1.683	1.861	0
2.00	5	120	0.793	0.866	0.937	1.024	1
2.00	10	120	0.897	0.972	1.046	1.130	0
2.00	20	120	0.922	1.013	1.087	1.161	2
2.00	50	120	0.938	1.018	1.111	1.169	1
2.00	100	120	0.952	1.031	1.152	1.208	0
2.00	200	120	0.967	1.066	1.181	1.260	0
2.00	500	120	0.958	1.064	1.285	1.394	0
2.00	1000	120	0.998	1.135	1.488	1.579	0
2.00	2000	120	1.138	1.294	1.917	2.115	0

Table S5: No-gate occupancy-ladder per-image fixed-dwell high-flux gain (per-image seed-mean radiometric PSNR at $\bar{\rho} = 0.6$ minus $\bar{\rho} = 0.05$, both at $\nu = 2000$, equal mean detector load and equal incident dose across occupancies). $k = 16$ is the gate-carrying arm (values are the frozen ΔQ_{budget}); $k = 512/32/1$ are no-gate controls/robustness rungs. The profile is nonmonotone in sparsity: $k = 32$ attains the largest median gain, combining most of the contrast benefit with half the conditioning burden of $k = 16$.

image	family	ΔPSNR (dB) by occupancy k			
		k=512 (50%)	k=32 (3.1%)	k=16 (1.6%)	k=1 (0.1%)
`detail32_glyph_0`	glyph	+0.21	+3.62	+1.11	-7.10
`detail32_glyph_1`	glyph	+2.25	+9.05	+0.80	+0.60
`detail32_glyph_2`	glyph	+0.26	+3.99	+0.90	-6.04
`detail32_glyph_3`	glyph	+1.94	+9.14	+0.70	+0.59
`detail32_chirp_0`	chirp	+0.12	+3.02	+2.30	+5.37
`detail32_chirp_1`	chirp	+0.18	+2.40	+0.35	+6.91
`detail32_chirp_2`	chirp	+0.28	+4.10	+2.88	+5.20
`detail32_chirp_3`	chirp	+0.18	+4.70	+2.14	+3.48
`detail32_spokes_0`	spokes	+0.64	+2.76	+1.21	+3.47
`detail32_spokes_1`	spokes	+0.65	+3.78	+1.46	+3.33
`detail32_spokes_2`	spokes	+0.65	+5.41	+4.38	+3.19
`detail32_spokes_3`	spokes	+0.68	+4.02	+1.40	+3.64
`detail32_maze_0`	maze	+1.01	+9.62	+7.62	+3.13
`detail32_maze_1`	maze	+0.53	+8.73	+3.14	+3.88
`detail32_maze_2`	maze	+1.29	+9.01	+9.70	+0.78
`detail32_maze_3`	maze	+2.39	+10.28	+8.24	+0.11
`detail32_contour_0`	contour	+1.21	+5.29	+1.01	+5.27
`detail32_contour_1`	contour	+1.40	+5.66	+0.02	+5.42
`detail32_contour_2`	contour	+1.25	+4.17	-0.32	+3.81
`detail32_contour_3`	contour	+1.41	+5.07	+1.35	-8.26
`detail32_microtexture_0`	microtexture	+0.12	+1.94	-0.98	+6.26
`detail32_microtexture_1`	microtexture	+0.06	+1.76	-1.64	+5.29
`detail32_microtexture_2`	microtexture	+0.15	+2.69	-0.11	+6.92
`detail32_microtexture_3`	microtexture	+0.21	+2.21	-1.49	+6.80
median		**+0.64**	**+4.13**	**+1.16**	**+3.56**
scenes > 0		24/24	24/24	19/24	21/24

Table S6: Dynamic-scattering detail: RQL radiometric PSNR under multiplicative lognormal AR(1) scattering ($\mathrm{CV}(\alpha) = 0.20$, applied before dead time) on the $k = 16$ geometry over the six frozen mechanism representatives. Entries are mean $\pm$ sample sd (dB) over the six scenes $\times$ three seeds; the gain column is the fast-minus-safe difference of means. The high-flux gain persists and grows with dwell.

ν	safe ($\bar{\rho}$=0.05)	fast ($\bar{\rho}$=0.6)	gain (dB)
20	8.71 ± 2.73	9.37 ± 2.75	+0.66
200	8.71 ± 2.87	10.55 ± 3.30	+1.84
2000	9.96 ± 3.60	12.28 ± 4.14	+2.32

Table S7: Undersampling sensitivity (S2B): median RQL radiometric PSNR (dB) for dense Bernoulli(1/2) illumination at two sampling budgets $M \in \{1024, 2048\}$, over 12 scenes $\times$ 3 seeds per cell. Source: `round63_s2_s2b/pilot_rows.csv` (commit 7b9d20e).

M	$\bar{\rho}$	ν=100	ν=500	ν=2000
1024	0.05	14.46	14.37	14.46
	0.60	14.34	14.57	15.01
	1.00	14.58	14.50	15.17
2048	0.05	14.38	14.53	14.58
	0.60	14.40	14.95	15.71
	1.00	14.58	14.70	15.89

Table S8: Ensemble sensitivity (S2C): median RQL radiometric PSNR (dB) for complementary Hadamard-type pairs (`hadpair`) and gamma-modulated patterns (`gam4`) at $M = 2048$, over 12 scenes $\times$ 3 seeds per cell. Source: `round63_s2_s2c/pilot_rows.csv` (commit UNTRACKED).

pattern	$\bar{\rho}$	ν=100	ν=500	ν=2000
	0.05	13.51	13.99	14.27
`hadpair`	0.60	13.89	14.63	15.85
	1.00	13.88	14.74	16.08
	0.05	13.50	13.46	13.57
`gam4`	0.60	13.46	13.52	13.94
	1.00	13.46	13.69	14.18